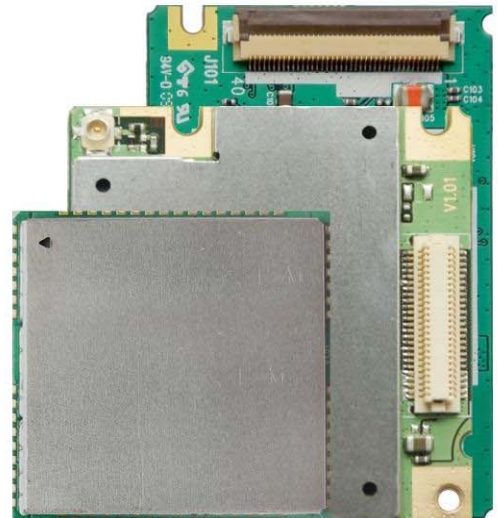




Quectel Cellular Engine

GSM FILE AT Commands

GSM_FILE_ATC_V1.2



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0. Revision history

Revision	Date	Author	Description of change
1.0	2010-04-15	Jay XIN	Initial
1.1	2010-06-10	Jay XIN, Joanna LI	Support SD and RAM
1.2	2010-10-08	Jay XIN	1. Add the ACK mode when uploading files 2. Modify the meaning of AT+QFUPL command parameters <timeout>

1. Introduction

Quectel Module provides AT commands to operate files in RAM, flash and SD card. This document is a reference guild to these commands.

1.1. Reference

Table 1: Reference

SN	Document name	Remark
[1]	Mxx_ATC	The introduction of AT commands for Mxx

1.2. Terms and abbreviations

Table 2: Terms and abbreviations

Abbreviation	Description
UFS file	File saved in User File Storage directory in Module flash
RAM file	File saved in RAM, only M33 supports
SD file	File saved in Picture directory of SD card, only M33 supports

2. AT Commands for FILE

Quectel Module provides AT commands to operate files in RAM, flash and SD card. Only M33 supports the file operations on RAM and SD card. And The SD card only supports three file systems: FAT, FAT16 and FAT32.

2.1. Overview of AT commands for File

Command	Description
<u>AT+QFLDS</u>	Get storage data size
<u>AT+QFLST</u>	List files
<u>AT+QFUPL</u>	Upload file to storage
<u>AT+QFDWL</u>	Download file from storage
<u>AT+QFDEL</u>	Delete file in storage
<u>AT+QFMOV</u>	Move file

2.2. Detailed descriptions of commands

2.2.1. AT+QFLDS Get storage data size

AT+QFLDS Get storage data size	
Test Command AT+QFLDS=?	Response OK
Write Command AT+QFLDS=<namepattern>	Response +QFLDS: <free size>,<total size>[,<maxalloc size>] OK
	Parameter <namepattern> pattern "UFS" UFS file in flash "RAM" RAM file (M33 support only) "SD" SD file (M33 support only) <free size> free data size in <namepattern> <total size> total data size in <namepattern> <maxalloc size> the maximum size which can be allocated. only valid for RAM file,
Execution Command AT+QFLDS	Response +QFLDS:<ufs file size>,<ufs file number>

	OK Returns the UFS information
	Parameter <ufs file size> The size in bytes of all files in UFS <ufs file number> The number of files in UFS
Reference	

2.2.2. AT+QFLST List files

AT+QFLST List files	
Test Command AT+QFLST=?	Response OK
Write Command AT+QFLST=<namepattern >	Response +QFLST: <file name>,<file size>[,<ram size>] OK
	Parameter < namepattern > pattern for filename “*” All UFS file in flash “RAM:*” All RAM file(M33 support only) “SD:*” All SD file (M33 support only) “filename” Name of UFS file “RAM:filename” Name of RAM file “SD:filename” Name of SD file <file name> Name of the file <file size> Size in bytes of the file <ram size> Memory size allocated for the file in RAM, only valid for RAM file
Execution Command AT+QFLST	Response +QFLST: <file name>,<file size> OK List files in the UFS directory
	Parameter <file name> Name of the file <file size> Size in bytes of the file
Reference	Note Only list files in the “Picture” directory of SD card, do not list any directory and any file in the other directories.

2.2.3. AT+QFUPL Upload file to storage

AT+QFUPL Upload file to storage	
Test Command AT+QFUPL=?	Response +QFUPL: "file name",(1-102400),(1-65535),(0,1) OK
	Parameter See Write Command.
Write Command AT+QFUPL=<file name>[,<file size> [,<timeout>[,<ackmode>]]]	Response CONNECT TA switches to data mode, and the bin data of file can be inputted. When the total size of the input data reaches <file size> (unit: byte) or TA receives “+++” sequence from UART, TA returns to command mode and replies the following codes. +QFUPL: <upload size>,<checksum> OK
	Parameter <file name> The name of the file to be stored. <file size> The maximum size of the file to upload. Default is 10240. Unit: byte <upload size> The size of the actually uploaded data. Unit: byte <timeout> The time in seconds to wait data input from UART. Default is 5. <ackmode> Whether to use acknowledge mode 0 Turn off the ACT mode. It's default 1 Turn on the ACT mode <checksum> The checksum of the uploaded data
Reference	Note: - It is strongly recommended to use DOS 8.3 file name format for <file name>. <file name> “filename” File is uploaded to the UFS directory “RAM:filename” File is uploaded to RAM (M33 support only) “SD:filename” File is uploaded to Picture directory in SD card (M33 support only) <checksum> is 16 bit checksum based on bitwise XOR. “+++” sequence will cause TA to end the command and switch to command mode; however, data previously entered are still being preserved as the data of the file.

	- To execute the command, must appear "CONNECT" before entering the binary data. - The ACT mode is provided to avoid loss of data when uploading large files in case hardware flow control is not available. The ACT mode works as follows: - Run command AT+QFUP = "file name", filesize, 5,1 to enable the ACK mode; - The module outputs "CONNECT"; - MCU sends 1Kbytes data, and then the module will respond with a 'A'; - MCU receives this 'A' and then sends the next 1Kbytes data - Repeat step 3) and 4) until the transfer is completed.
--	---

2.2.4. AT+QFDWL Download file from storage

AT+QFDWL Download file from storage	
Test Command AT+QFDWL=?	Response +QFDWL: "file name" OK
	Parameter See Write Command.
Write Command AT+QFDWL=<file name>	Response CONNECT TA switches to data mode, and the bin data of the file will be outputted. When the file was read over, TA returns to command mode and replies the following codes: +QFDWL: <download size>,<checksum> OK
	Parameter <file name> The name of the file to be downloaded <download size> The size of the downloaded data <checksum> The checksum of the downloaded data
Reference	Note: <file name> "filename" File is uploaded to the UFS directory "RAM:filename" File is uploaded to RAM (M33 support only) "SD:filename" File is uploaded to Picture directory in SD card (M33 support only)

	“+++” sequence will cause TA to end the command and switch to command mode. <checksum> is 16 bit checksum based on bitwise XOR.
--	--

2.2.5. AT+QFDEL Delete file in storage

AT+QFDEL Delete file in storage	
Test Command AT+QFDEL=?	Response +QFDEL: "file name"
	OK
	Parameter See Write Command.
Write Command AT+QFDEL=<file name>	Response OK
	Parameter <file name> The name of the file to be deleted “*” Delete all files in UFS directory (not delete the directory) “RAM:*” Delete all files in RAM (M33 support only) “SD:*” Delete all files in Picture directory of SD card (M33 support only) “filename” delete the specific file “filename” in UFS directory “RAM:filename” delete the specific file “filename” in RAM (M33 support only) “SD:filename” delete the specific file “filename” in Picture directory of SD card (M33 support only)
Reference	Note Only delete file in the Picture directory of SD card, do not delete any directory and any file in the other directories.

2.2.6. AT+QFMOV Move file

AT+QFMOV Move file (M33 support only)	
Test Command AT+QFMOV=?	Response +QFMOV: "src filename", "dest filename", (0,1), (0,1)
	OK
	Parameter

	See Write Command.
Write Command AT+QFMOV=<src filename>,<dest filename>,<copy>,<overwrite> >	Response OK Parameter <src filename> source file <dest filename> destination file <copy> whether deleting source file after the file is copied 0 delete source file after file is copied 1 not delete source file after file is copied <overwrite> whether overwrite existed destination file 0 Do not overwrite the destination file if it exists 1 Overwrite the destination file if it exists
Reference	Note: ● This command is supported only in M33. ● Can not move file from UFS or SD card to RAM. ● AT+QFMOV="RAM:*","SD:*",1,1 Move all files in RAM to SD card ● AT+QFMOV="RAM:filenamea","SD:filenameb",1,1 Move the file named "filenamea" in RAM to SD card, and rename it as "filenameb".

3. Summary of error codes

Final result code **+CME ERROR: <err>** indicates an error related to mobile equipment or network. The operation is similar to **ERROR** result code. Neither **ERROR** nor **OK** result code shall be returned. The listed **<err>** codes here are just related with File. About other **<err>** codes, please refer to document [1].

Code of <err>	Meaning
3915	Non-existent address
3916	UFS storage full
3917	Drive full
3918	Drive error
3919	File not found
3920	Invalid file name
3921	File already existed
3922	Failed to create file
3923	Failed to write file
3924	Failed to open file
3925	Failed to read file

4. Reliable transmission

4.1. Calculated checksum to check the file transfer

For reliable transmission when using "AT+QFUPL" and "AT+QFDWL" to upload and download file, it's recommended that users turn on hardware flow control capabilities, while also opening MCU hardware flow control function. Using the **AT+IFC=2,2<CRLF>** command to open the hardware flow control function for the module. It is turned on by default.

As general serial transmission is reliable, in order to further reliability, we provide additional ways to verify the data transmission reliability by the command's response information.

When using "AT+QFUPL=<file name> [,<file size>]" command to upload a file, the module will report "+QFUPL: <upload size>, <checksum>" information tips at the end of data transmission. Then MCU can judge whether the data has lost by comparing value of <upload size> and <checksum>.

<upload size> is the data length which the module received. MCU compares <upload size> and the actual length of the file. If unequal, it means the module lost data.

<checksum> is calculated by doing XOR for every 2 bytes. Similarly MCU calculates the actual file's checksum as below example, and then compares this value with <checksum> which module reports. If not equal, the received data may be problematic. User can re-upload data.

Example for calculating checksum:

If the uploaded file data length is 9, the 16 hex values are as follows:

0x23 0x13 0x65 0x B6 0x76 0x88 0xA3 0xEF 0x55

So, checksum is calculated as follows:

checksum = 0x2313 XOR 0x65B6 XOR 0x7688 XOR 0xA3EF XOR 0x5500

Every two data form a group and do XOR with another group. If the last group is less than 2 bytes, supplement with 0x00.

Similarly, the module will report the "+QFDWL: <download size>, <checksum>" information when command "AT+QFDWL=<file name>" is completed. <download size> is the actual size of downloaded data, MCU can calculate received data length, and compare it with <download size>. If not equal, the data is lost. MCU also can do checksum calculation and comparison with <checksum>, if not equal, need to re-download.

4.2. the ACK mode is enabled to stabilize Uploading

When you need to use the AT + QFUPL command to upload large files, and UART hardware flow control is turned off, it's recommended to use the ACK Mode.

The ACT mode works as follows:

- 1) Run command AT+QFUPL = "file name", filesize, 5,1 to enable the ACK mode;
- 2) The module outputs "CONNECT";
- 3) MCU sends 1Kbytes data, and then the module will respond with a 'A';
- 4) MCU receives this 'A' and then sends the next 1Kbytes data
- 5) Repeat step 3) and 4) until the transfer is completed.

For example:

```

AT+QFUPL="test.txt",3000           // Upload the text file "test.txt" to UFS
CONNECT
<input file bin data of 1024bytes>
A                                   //After receive 1024bytes data, module will response
                                   with a "A", then next 1024 bytes data can be input
<input file bin data of 1024bytes>
A
<input the rest file bin data>
+QFUPL: 3000,B34A
OK
  
```

5. Examples

5.1. File uploading and downloading

```

AT+QFUPL="test.txt",3222           // Upload the text file "test.txt" to UFS
CONNECT
<input file bin data>
+QFUPL: 3222,B3E4
OK

AT+QFDWL="test.txt"               // Download the file "test.txt" from UFS
CONNECT
<output file bin data>
  
```

```
+QFDWL: 3222,B3E4
```

```
OK
```

```
AT+QFUPL="RAM:test2.txt",4222
```

```
// Upload the text file "test2.txt" to RAM
```

```
CONNECT
```

```
<input file bin data>
```

```
+QFUPL: 4222,13E4
```

```
OK
```

```
AT+QFDWL="SD:pic1.jpg",13222
```

```
//Download the picture file "pic1.jpg" from SD card
```

```
CONNECT
```

```
<input file bin data>
```

```
+QFUPL: 13222,D5E4
```

```
OK
```

5.2. File moving

User can move file(s) among RAM, UFS and SD card by command "AT+QFMOV". Please remember user can not move file from UFS or SD card to RAM. Here list examples of moving single file and all files.

5.2.1. Move single file

```
AT+QFLST="RAM:*" //RAM has file "Pic.jpg"
```

```
+QFLST: "RAM:Pic.jpg",63388,75000
```

```
OK
```

```
AT+QFLST="*" //UFS also has one file "pic.jpg", but the file size is different
```

```
+QFLST: "pic.jpg",62076
```

```
OK
```

```
AT+QFMOV="RAM:pic.jpg","pic.jpg",1,0
```

```
+CME ERROR: 3921
```

```
//move file "pic.jpg" from RAM to UFS, "1" means
```

```
NOT deleting source file after file is copied,
```

```
"0" means "Do not overwrite the destination
```

```
file if it exists". Because the destination file has existed,
so it responses ERROR 3921 (File already existed).
```

```
The file is not moved. If user confirms the file can be
overwritten, set the last parameter as "1" as below.
```

```
AT+QFMOV="RAM:pic.jpg","pic.jpg",1,1
```

```
//Moving file successfully. The source file
```

“pic.jpg” in RAM is not deleted. The destination file “pic.jpg” in UFS has been overwritten,

OK

AT+QFLST="RAM:*"

+QFLST: "RAM:Pic.jpg",63388,75000

OK

AT+QFLST="*"

+QFLST: "pic.jpg",63388

OK

5.2.2. Move all files of one storage

AT+QFLST="RAM:*"

+QFLST: "RAM:pic0.jpg",59024,75000

+QFLST: "RAM:pic1.jpg",62592,75000

+QFLST: "RAM:pic2.jpg",57168,75000

+QFLST: "RAM:pic3.jpg",63216,75000

+QFLST: "RAM:pic4.jpg",64600,75000

+QFLST: "RAM:pic5.jpg",60284,75000

OK

AT+QFLST="SD:*"

OK

AT+QFMV="RAM:","SD:",0,0

OK

//Move all files in RAM to SD, the first “0” means deleting source files after files are copied, the second “0” means “Do not overwrite the destination file if it exists”. SD has not these files, so the last parameter is meaningless in this case.

AT+QFLST="RAM:*"

//Source files are deleted

OK

AT+QFLST="SD:*"

//All files in RAM have been moved to SD card

+QFLST: "SD:pic0.jpg",59024

+QFLST: "SD:pic1.jpg",62592

+QFLST: "SD:pic2.jpg",57168

+QFLST: "SD:pic3.jpg",63216

+QFLST: "SD:pic4.jpg",64600

+QFLST: "SD:pic5.jpg",60284

OK



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